Open Questions in the Evolution of the Bacterial Flagellum

A corpus-grounded synthesis

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Abstract

The bacterial flagellum is among the most-studied molecular machines, yet its evolutionary history remains only partially resolved. Decades of structural, genetic, and comparative work have converged on a clear picture of *what* the flagellum is and *how* it works, while leaving several *how-did-it-get-here* questions genuinely open. This note frames four such questions that the underlying literature corpus itself surfaces—the direction of evolution between the flagellar export apparatus and the type III secretion system (T3SS), the origin and diversification of the ion-powered motor, the relationship between the bacterial flagellum and the archaeal archaellum, and the reconstruction of a minimal ancestral machine—and summarizes the specific evidence and the specific disagreements around each. All claims are drawn from a 516-paper flagellum/ATP corpus and retrieved via a local retrieval-augmented-generation service.

Contents

1 Introduction

The flagellum is a supramolecular motility machine built from roughly 30 different proteins, organized into a basal-body rotary motor, a hook acting as a universal joint, and a helical filament that serves as a propeller [15]. Estimates of the parts count vary with how one counts: at least 40 genes are involved, with at least 24 contributing proteins to the final structure [4], alongside an earlier figure of ∼20 structural proteins plus ∼30 more for construction, function, and maintenance [11]. This complexity—a self-assembling, ion-powered, reversible rotary motor—is precisely what makes its evolution scientifically interesting.

Importantly, the mechanistic maturity of the field does not translate into a settled evolutionary account. The corpus is dense with structure and mechanism and comparatively thin on explicit macro-evolutionary modeling; where it does address origins, it repeatedly surfaces the same handful of unresolved questions. We treat those as the paper's organizing spine.

2 Which came first — the flagellum or the T3SS?

The single most-developed evolutionary thread in the corpus is the deep relationship between the flagellar export apparatus and the virulence-associated T3SS. The two are "evolutionarily and functionally related" [5]: roughly half of the ∼20–25 T3SS proteins are conserved across type III systems, and most are also similar in sequence to basal-body proteins of the flagellum [6]; up to eight flagellar-assembly gene products are homologous to conserved T3SS components [14]. Macnab put it most strongly—the flagellar pathway *is* a type III pathway, "differing only in the nature of its export substrates and in the fact that it operates via a working organelle of propulsion" [9].

Consensus vs. the live disagreement

The preponderance of evidence favors the flagellum as the **ancestral** system, with the virulence T3SS (Type IIIb) derived from it: flagella are ancient and predate metazoan/plant hosts [14, 9]; flagellar (Type IIIa) homologs are more similar to one another than to virulence components [14]; and Type IIIb systems are patchily distributed on plasmids/pathogenicity islands, implying later horizontal transfer [14]. *But* a dissenting analysis holds the T3SS is as ancient as the flagellar apparatus and that the two share a **common ancestor** rather than one descending from the other [6]—a genuine topological disagreement the corpus cannot yet adjudicate.

3 How did the ion-powered motor originate and diversify?

The torque-generating core is strikingly conserved: five proteins—the stator (MotA/MotB or homologs) and rotor proteins FliG, FliM, FliN—recur across motor types, and the FliG C-terminal domain is largely interchangeable between species [18]. Chimeric motors that mix proton- and sodium-type components function [12], arguing for a common ancestral mechanism that diversified mainly in ion coupling.

3.1 A concrete open puzzle: the selectivity filter

A single aspartate in MotB/PomB (D32 in *E. coli* MotB [3]; D24 in PomB [7]) is essential and universally conserved across stator families—which means it *cannot* distinguish a sodium motor from a proton motor [7]. Recent cryo-EM of the *Vibrio* PomAB complex located Na^+ selectivity partly in three threonines of PomA (T158/T185/T186), conserved in all sodium stators [7], revising the older view that the B subunit alone sets specificity. How selectivity is encoded—and therefore how it *changes* over evolution—is an active, unfinished structural problem.

3.2 Diversification is real

Stator systems span single-ion (MotAB/PomAB), multiple-stator genomes ($\geq$65 species carry two or more stators; *V. alginolyticus* runs Na^+ PomAB polar and H^+ MotAB lateral systems), and genuinely dual-ion stators (*Bacillus clausii* MotAB couples H^+ or Na^+; *B. alcalophilus* MotPS couples $Na^+/K^+/Rb^+$) [3]. Some lineages bolt on extra parts—the *Vibrio* Na^+ motor requires outer-membrane MotX/MotY that engineered hybrids do without [18, 8]. The open question is the *order and drivers* of this diversification: which coupling ion is ancestral, and how many times has switching occurred? An energy-transduction mechanism shared across export and rotation further constrains plausible intermediates [10].

4 Flagellum vs. archaellum — one origin or two?

Comparative data force a sharp conclusion. The archaeal motility organelle (archaellum) does the same job by entirely different means: archaeal genomes contain **no homologs** of bacterial flagellins, rod, hook, ring, switch, or Mot proteins [16]. Archaeal filaments are thinner (10–14 nm vs. $\sim$20 nm), built from multiple glycosylated flagellins made with cleaved leader peptides, and appear to lack the central assembly channel that defines bacterial flagellar growth [16, 17]. Instead, archaeal flagellins resemble **type IV pilins**, and archaellum assembly factors resemble type IV pilus NTPases and membrane proteins [17].

The corpus's own reading is that flagellar biogenesis "may have evolved independently in these two domains of life" [17]—convergent evolution of rotary swimming from unrelated parts. A telling wrinkle: archaea retain bacterial-like chemotaxis genes (*cheA*/*cheW*/*cheY*) bolted onto a non-homologous motor [16], so the *control system* is shared while the *machine* is not.

5 Can we reconstruct a minimal ancestral machine?

The T3SS thread offers the corpus's clearest stab at a stepwise origin. The observation that ATPase activity is *dispensable* for type III export [5] reframes the classic "flagellum from a proto-F_oF_1 ATP synthase" story: rather than ATP hydrolysis being foundational, a **proton-powered primordial export system** may have come first, with a proto-F_1-ATPase *added later* to facilitate export [5]. That inverts the intuitive order (energy module first) and suggests a plausible ancestral intermediate—a secretion pore predating both motor and ATPase. But the corpus carries no full ancestral-state reconstruction, and is candid that it lacks substantive treatment of the irreducible-complexity debate, pointing outward to work not in the set [15]. The open question is empirical and tractable: what is the smallest functional sub-assembly that is both selectable and on a credible path to the modern motor?

6 Synthesis: a conserved core, an unconserved history

Across all four questions a single pattern recurs—a **strongly conserved functional core** (rotor–stator electrostatics [2, 18], the export apparatus, the three-part body plan) surrounded by **lineage-specific elaboration** (divergent peripheral motor structures seen by electron cryotomography [15], extra stator parts, alternate ion couplings). The machine is conserved; its *history* is not agreed. The productive open questions are less "could this evolve?" and more "in what order, powered by which ion, and from which pre-existing pore?"—answerable with phylogenomics and ancestral-sequence reconstruction, methods largely outside the current corpus, which defines the most useful direction for augmenting it.

This synthesis was generated by retrieving passages from a 516-paper flagellum/ATP corpus via a local retrieval-augmented-generation service and synthesizing with an LLM; every claim traces to a retrieved passage. The reference list was reconstructed by extracting bibliographic metadata directly from the parsed source PDFs. Two references warrant note: the corpus's internal key `Sowa-2008` in fact resolves to Biquet-Bisquert et al. (2021), and `Sowa-2008-b` to Terashima et al. (2017)—the automated author-year keying during corpus assembly produced collisions that have been corrected here against the actual PDFs. No bibliographic detail was invented; fields that could not be verified from a source were left blank.

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